MPTA-Repair: Clock-Bound Repair for Multi-Property Timed Automata in Industrial Practice

***Abstract*—Timed automata (TAs) are widely used to model and verify real-time industrial systems, which are typically required to satisfy multiple predefined properties simultaneously. When verification fails, repairing the model is challenging because it requires modifying erroneous clock constraints to satisfy all properties while preserving functional equivalence. To address this, we present MPTA-Repair, an automated repair framework for multi-property TAs. The framework employs an iterative, counterexample-guided MaxSMT approach to generate minimally modified candidate models. To optimize search efficiency, it systematically exploits relationships among properties, counterexamples, and repair variables. Each candidate model is then validated through full-property regression verification and acceptability checks. Compared to the TARTAR baseline, MPTA-Repair achieves higher effectiveness on faulty models generated via our proposed mutation strategy. Specifically, it improves the success rate from 72% to 93% on benchmarks derived from UPPAAL and the XtaBenchmarkSuite, and from 20% to 76% on an industrial dataset constructed in this work.**

***Index Terms*—Timed automata, automated model repair, MaxSMT, multi-property verification**

I. Introduction

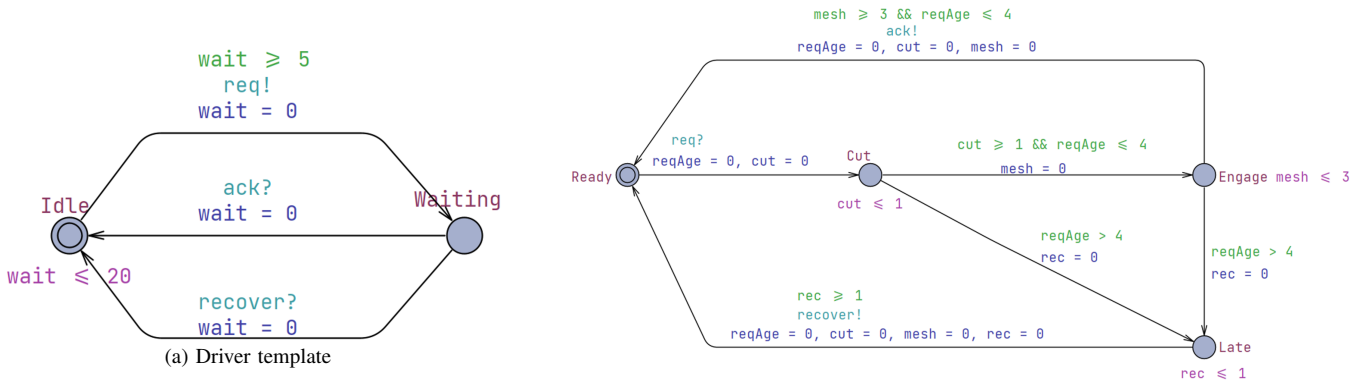
Timed automata (TAs) are widely used to model and verify real-time industrial systems whose correctness depends on both discrete behavior and quantitative timing constraints [1], [2]. Such systems commonly appear in railway control and communication protocols [3], automotive and autonomous-driving logic [4], [5], medical cyber-physical systems [6], [7], and aerospace or avionics scheduling [8]. In these domains, an alarm, pulse, message, or resource release may be functionally present but still incorrect if it occurs too early, too late, or for an incorrect duration. Verification therefore has to check not only reachability or event order, but also whether timing requirements are satisfied under all relevant executions.

Industrial timed-automata models are rarely validated against a single property. They are typically checked against a set of predefined properties that jointly express safety, bounded response, deadline satisfaction, error-state reachability, alarm duration, and deadlock freedom [9]. When a property is violated, model checkers such as UPPAAL, Kronos, or opaal can produce a timed diagnostic trace (TDT), namely a timed counterexample showing how the model reaches a violating state [10]. A TDT is useful evidence for repair, but it is not a repair instruction. It does not identify which clock constraint is erroneous, whether the error lies in a transition guard or a location invariant, or how a numerical bound should be changed to restore the intended timing behavior [11].

The difficulty becomes more pronounced in the multi-property setting. A timing fault may affect several properties, and different violated properties may produce related counterexamples. A repair that blocks one reported trace may leave another violating trace feasible, or may break a property that was previously satisfied. Similarly, changing a bound for one property can affect another property when the same clock, location, transition, synchronization, or execution fragment is involved. Thus, a practical repair method for industrial timed automata must not accept a candidate merely because it fixes one local counterexample. It must validate the candidate against the complete property set and ensure that the functional behavior of the model is preserved.

Figure 1 illustrates this issue with a simplified network timed automaton for an automotive gear-shift controller [4]. A Driver template periodically issues a shift request, and a GearController template performs torque cut, gear engagement, acknowledgement, and recovery. The model uses $cut \leq 1$ in the Cut location, $cut \geq 1$ on the transition to Engage, $mesh \leq 3$ in the Engage location, and $mesh \geq 3$ on the acknowledgement transition. One property requires at least two time units of torque cut before gear engagement, while another property requires the shift request to be acknowledged within four time units. A single-property repair for the first property may change only the cut bounds from 1 to 2 and thereby block the early-engagement trace. However, with the mesh bounds unchanged at 3, the earliest acknowledgement would occur at time 5, so the response-deadline property fails. A multi-property repair instead changes both the cut bounds and the mesh bounds, for example to $cut \leq 2$, $cut \geq 2$, $mesh \leq 2$, and $mesh \geq 2$. The repaired controller then satisfies both the minimum torque-cut requirement and the response deadline while preserving the same untimed request-acknowledgement behavior. This example motivates treating properties, counterexamples, and repair variables as joint repair evidence rather than independent local repair tasks.

Existing repair techniques provide an important foundation for this problem. Clock-bound repair encodes a TDT as linear real arithmetic, introduces variation variables for clock bounds, and uses MaxSMT solving to compute small modifications to guards and invariants [11]. TARTAR builds on this idea by suggesting syntactic repairs and checking whether the repaired model preserves the untimed functional behavior of the original network timed automaton [12]. Related work also includes parameter synthesis for parametric timed automata



(a) Driver template

(b) GearController template

Fig. 1. A simplified network timed automaton for a cyclic automotive gear-shift controller. The faulty model uses $cut \leq 1$, $cut \geq 1$, $mesh \leq 3$, and $mesh \geq 3$. A repair that changes only the cut bounds blocks the early-engagement trace but can violate the response-deadline property; a multi-property repair also changes the mesh bounds.

[13], clock-guard repair through abstraction and testing [14], robustness analysis for uncertain timing constants [15], and SAT/SMT-based model or program repair [16]. These methods show that automated repair of timed models is feasible, but most TDT-based workflows still reason from one violated property or one diagnostic trace at a time.

This paper presents **MPTA-Repair**, an automated repair framework for multi-property TAs. Given a faulty timed-automata model, a set of timing properties, and diagnostic traces produced by verification, MPTA-Repair searches for small changes to numerical clock bounds in transition guards and location invariants. It does not change property formulas, locations, transitions, synchronization labels, clock references, or resets. This restriction matches the clock-bound repair formulation and the mutation-based evaluation used in this paper, while broader modeling faults such as reset errors, synchronization errors, data-update errors, or structural errors are outside the current scope.

MPTA-Repair employs an iterative, counterexample-guided MaxSMT approach to generate minimally modified candidate models. Instead of treating violated properties and diagnostic traces as independent subproblems, it maintains a pool of property-trace repair obligations and searches for a clock-bound assignment that satisfies the currently encoded obligations. To improve search efficiency, the framework systematically exploits relationships among properties, counterexamples, and repair variables to organize the search. These relations group related evidence, rank or restrict active repair variables, and avoid solving over unrelated bounds when the current counterexample evidence gives no reason to include them. These relations guide candidate generation, whereas accepted candidates must still satisfy all properties in regression verification and pass a TARTAR-style acceptability check based on functional equivalence.

We evaluate MPTA-Repair on faulty models generated via a mutation strategy designed for clock-bound repair. The benchmark dataset is derived from UPPAAL and the XtaBenchmarkSuite, while the industrial dataset is constructed in this work from practical case studies covering railway alarm and communication scenarios [3], [17], train-gate behavior [18], automotive gear control [4], cardiac pacing [6], autonomous road-junction rules [5], mechanical ventilation [7], and aerospace scheduling [8]. We measure repair effectiveness, repair size, runtime, regression outcomes, and behavior preservation. Compared with an iterative TARTAR-style baseline under the same repair space, full-property regression requirement, and admissibility requirement, MPTA-Repair improves the repair success rate from 72% to 93% on the benchmark dataset and from 20% to 76% on the industrial dataset.

The contributions of this paper are as follows. First, we formulate clock-bound repair for practical timed-automata models as a multi-property, multi-counterexample repair problem in which local trace repair is insufficient unless the repaired model satisfies the complete property set. Second, we present MPTA-Repair, an iterative counterexample-guided MaxSMT framework that uses relationships among properties, counterexamples, and repair variables to guide the repair search. Third, we implement a prototype that combines trace analysis, MaxSMT-based clock-bound repair, full-property regression verification, and TARTAR-style admissibility checking. Fourth, we evaluate the approach on benchmark and industrial datasets against an iterative single-property baseline under identical acceptance criteria. Finally, we report practical lessons on property selection, repair interpretability, counterexample accumulation, and the need for full regression after each candidate repair.

The remainder of this paper is organized as follows. Section II provides the background and problem setting, including timed automata, diagnostic traces, clock-bound repair, and the multi-property repair objective. Section III presents MPTA-Repair and its relation-guided repair workflow. Section IV presents the implementation and experimental evaluation, including datasets, results, and practical lessons. Section V discusses threats to validity. Section VI reviews related work. Section VII concludes the paper.

II. Background and Motivation

A. Timed Automata and Industrial Networks

Timed automata provide a standard formal model for systems whose correctness depends on both discrete control decisions and quantitative time constraints [1], [2]. Let C be a finite set of clocks. A clock constraint over C is a conjunction of atomic constraints of the form $x \sim b$, where $x \in C$, $\sim \in \{<, \leq, =, \geq, >\}$, and b is a non-negative numerical bound. We write $\mathcal{B}(C)$ for the set of such constraints. A timed automaton is a tuple

$$T = (L, \ell_0, C, \Sigma, \Theta, I),$$

where L is a finite set of locations, $\ell_0 \in L$ is the initial location, Σ is a set of action labels, Θ is a finite set of transitions, and $I : L \to \mathcal{B}(C)$ assigns an invariant to each location. A transition $\theta \in \Theta$ is written as

$$\theta = (\ell, g, a, R, \ell'),$$

where ℓ and ℓ' are the source and target locations, $g \in \mathcal{B}(C)$ is the guard, $a \in \Sigma$ is the action label, and $R \subseteq C$ is the set of clocks reset by the transition.

The semantics is defined over states (ℓ, u), where ℓ is a location and $u : C \to \mathbb{R}_{\geq 0}$ is a clock valuation. A delay step lets time elapse in the current location as long as the location invariant remains satisfied. A discrete step can be taken when the current clock valuation satisfies the transition guard; the step moves the automaton to the target location and resets the clocks in R. Thus, guards constrain when discrete actions are enabled, whereas invariants constrain how long the automaton may remain in a location.

Industrial timed-automata models are usually networks rather than single automata. A network

$$\mathcal{N} = T_1 \parallel \cdots \parallel T_k$$

composes templates for controllers, plants, environments, communication protocols, schedulers, and monitors [19]. The exact product semantics depends on the modeling language and tool, for example on binary synchronization, broadcast channels, urgent or committed locations, and data variables. In this work these structural and data-level features are treated as fixed modeling context. The repair space is deliberately restricted to numerical clock-bound occurrences in guards and invariants; synchronization labels, reset sets, clock references, locations, transitions, data updates, and property formulas are not repair variables.

This restriction matches a common class of industrial modeling faults. Timing constants often represent design thresholds, sampling periods, timeout values, alarm durations, refractory intervals, or scheduling deadlines. For example, vehicle controllers may need to complete a gear-shifting sequence within a bounded response time [4]; railway models may encode gate, alarm, or communication timeout requirements [3]; medical devices may encode physiological timing windows [6]; and avionics or aerospace models may encode execution-time bounds and deadlines [20]. In such models, an incorrect numerical bound can make an otherwise intended behavior occur too early, too late, or for an invalid duration.

We do not assume that all modeling errors are clock-bound errors. Industrial models may also contain wrong resets, missing transitions, incorrect synchronizations, erroneous data updates, or incomplete properties. These faults are outside the current repair space. The assumption in this paper is narrower: the discrete model structure is treated as the intended design, while one or more numerical timing bounds may be inaccurate due to calibration, transcription, dimensioning, or design-space uncertainty. This is the fault model targeted by clock-bound repair [11] and is consistent with mutation-based repair and testing, where faults are modeled as small deviations from an otherwise near-correct artifact [21].

B. Properties and Timed Diagnostic Traces

Timed-automata models are checked against timing properties using real-time model checkers such as UPPAAL [19]. This paper focuses on safety-style verification obligations. Direct examples include queries of the form $A[]\,\psi$, where every reachable state must satisfy the state predicate ψ, and deadlock-freedom checks such as $A[]\ not\ deadlock$. Bounded-response and deadline requirements are handled when they are encoded by observer or monitor templates into safety-style violations, for example by reaching an error or late location [9]. Therefore, the repair procedure reasons about property violations that can be represented as reachable safety violations in the instrumented timed-automata network.

A discrete trace skeleton is a finite sequence of transitions

$$\tau = \theta_0, \theta_1, \ldots, \theta_{n-1}.$$

A timed realization of τ is a sequence of non-negative delays and clock valuations that makes the transition sequence executable from the initial state while respecting guards, resets, invariants, and network synchronization. The skeleton τ is feasible if it has at least one timed realization. Given a property φ, a feasible skeleton is a timed diagnostic trace (TDT) for φ if at least one of its timed realizations reaches a state that violates the safety condition associated with φ [11]. Depending on the model checker, the reported counterexample may contain concrete delays, symbolic zones, or only enough information to reconstruct the discrete path and timing constraints.

A TDT is evidence of failure, not a root-cause explanation. It shows that some timing realization of some discrete behavior violates a property, but it does not identify which numerical bound is wrong, whether the fault lies in a guard or invariant, or how the model should be changed. Several bounds may jointly enable the violating realization, and one incorrect bound may contribute to several diagnostic traces. Conversely, blocking the discrete path of a TDT is usually not a valid repair: it may remove intended functional behavior, such as a request, acknowledgement, actuation, communication step, or scheduling decision. A useful repair should change the timing envelope of the behavior while preserving the relevant untimed functionality of the model.

C. Clock-Bound Repair and Admissibility

A repairable site is an occurrence of a numerical bound b in an atomic clock constraint $x \sim b$ that appears in a transition guard or a location invariant. A clock-bound repair changes this occurrence to a new value b', equivalently by introducing a variation variable v and using the repaired bound $b + v$. The comparison operator, clock reference, transition structure, location structure, synchronization label, and reset set are kept fixed. Bounds are constrained to remain in the numerical domain supported by the target model representation.

TDT-based clock-bound repair encodes the feasibility of a diagnostic trace as constraints over delays and clock values, then introduces variation variables for the bounds that occur in the encoded guards and invariants [11]. MaxSMT solving can then be used to prefer small repairs, for example by maximizing soft constraints of the form $v = 0$ and, when needed, minimizing the magnitude of nonzero variations [22]. In the single-trace setting, the resulting candidate is local: the same discrete trace skeleton should remain feasible, but no feasible realization of that skeleton should still witness the considered property violation.

Local trace repair is not sufficient for model repair. A bound change that eliminates one violating realization may create a new violation on another execution, fail another property, or make intended functionality unavailable. MPTA-Repair therefore treats MaxSMT-based trace repair as candidate generation only. A candidate is accepted only after two global checks. First, the repaired model must satisfy the complete property set under regression verification. Second, it must pass an admissibility check based on TARTAR-style functional equivalence [11].

The admissibility check compares untimed functional behavior rather than timed language equality. This distinction is necessary because a successful timing repair is expected to remove some violating timed realizations. Let Σ_f denote the functional alphabet used for admissibility, after excluding monitor-only, observer-only, and internal labels when appropriate. The intended criterion is

$$\mathrm{Untimed}_{\Sigma_f}(M_{bad}) = \mathrm{Untimed}_{\Sigma_f}(M_{rep}),$$

meaning that the repaired model should not add or remove functional action traces, although the timing of those traces may change. This criterion prevents repairs that satisfy safety properties merely by disabling intended behavior.

D. Why Multi-Property Repair Is Needed

Industrial timed-automata models are normally validated by a set of properties rather than by a single assertion [9]. The properties jointly describe the reliability envelope of the model, including absence of unsafe states, bounded response, timeout handling, deadline satisfaction, admissible residence times, and deadlock freedom. Let M_{bad} be a faulty timed-automata network and let

$$\Phi = \{\varphi_1, \ldots, \varphi_m\}$$

be the property set used to validate it. The goal is to construct a repaired model M_{rep} by changing a small number of clock-bound constants in guards and invariants such that:

$$M_{rep} \models \Phi,$$

the repaired bounds are well formed, and the repaired model preserves the projected untimed functional behavior of M_{bad}.

The need for joint reasoning can be seen in the gear-shift controller used as a running example. The driver periodically issues a shift request, and the controller performs torque cut, gear engagement, acknowledgement, and recovery. Suppose the faulty controller uses bounds such as $cut \leq 1$, $cut \geq 1$, $mesh \leq 3$, and $mesh \geq 3$. One property may require at least two time units of torque cut before gear engagement, while another property may require acknowledgement within four time units of the request. A single-property repair that changes only the cut bounds from 1 to 2 can block the early-engagement violation. However, if the $mesh$ bounds remain at 3, the earliest acknowledgement may occur at time 5, thereby violating the response deadline. A multi-property repair must instead consider both requirements and may need to change both the cut and $mesh$ timing bounds while preserving the same untimed request–engage–acknowledge behavior.

This example illustrates why violated properties, diagnostic traces, and repair variables should not be treated as unrelated subproblems. Different properties may share clocks, locations, synchronizations, or execution fragments. Multiple TDTs may expose different realizations of the same underlying timing defect. Conversely, a repair computed from one trace can be locally valid yet globally unacceptable after full regression. MPTA-Repair addresses this setting by accumulating multi-property and multi-counterexample repair evidence, generating small clock-bound candidates with MaxSMT, and accepting a candidate only after complete property regression and functional-admissibility checking.

III. Approach

A. Overview

MPTA-Repair is a counterexample-guided repair workflow for multi-property, multi-counterexample clock-bound repair of industrial timed-automata models [23]. Its input is a faulty model `M`, a property set Φ = {φ1, ..., φm}, and configuration parameters such as the maximum number of refinement rounds, candidate limits, and a global timeout. The repairable bounds are not given as a separate user input; they are extracted from the numerical clock bounds occurring in transition guards and location invariants [11]. The output is either a repaired model `M′` together with a set of clock-bound edits, or a failure report explaining why no admissible full-property repair was found within the configured budget.

The workflow is designed around two principles. First, candidate repair generation may be driven by local diagnostic traces, but candidate acceptance must be global. A candidate that repairs one violated property may cause another property to fail, because the same clock, location, transition, or synchronization may participate in several timing

requirements. MPTA-Repair therefore keeps collecting counterexamples from newly violated properties and uses them to further restrict the set of admissible repair candidates. This refinement continues until a candidate satisfies the complete property set and passes admissibility checking, or until the configured budget is exhausted. In our evaluation, we use a fixed timeout, for example 10 minutes per repair instance, to avoid unbounded refinement [24].

Second, multiple properties and multiple diagnostic traces are not treated as unrelated repair subproblems. A single faulty timing bound may contribute to several violations, and several traces may expose different timing realizations of the same underlying defect. MPTA-Repair therefore maintains a pool of property-trace obligations and searches for a clock-bound assignment that satisfies all currently encoded obligations. Relation analysis among properties, traces, and repairable bounds is used to reduce redundant work and guide the search, but it is not used as a substitute for verification [23].

At a high level, MPTA-Repair proceeds as follows. It verifies the model against all properties, collects TDTs for violated properties, extracts repairable clock-bound values, and builds a joint MaxSMT repair problem from the current property-trace pool. The resulting assignment is instantiated into a candidate model. The candidate is then checked against the complete property set and against a TARTAR-style admissibility criterion [11]. If validation fails, the rejected assignment is blocked, newly observed diagnostic traces are added, and the repair loop continues.

B. Bound Variation and Joint Trace Encoding

MPTA-Repair follows the bound-variation view used in clock-bound repair [11]. Let a repairable clock-bound value be a numerical bound β occurring in an atomic clock constraint $\mathtt{x} \sim \beta$, where $\mathtt{x}$ is a clock and $\sim \in \{<, \leq, =, \geq, >\}$ is fixed by the original model. For each such bound, MPTA-Repair introduces a bound variation variable $\mathtt{v}$. The repaired constraint is written as:

$\mathtt{x} \sim (\beta + \mathtt{v})$.

The current repair pipeline changes only the numerical bound. The clock reference, comparison operator, transition structure, location structure, synchronization labels, and reset set are not repair variables in the main experiment. Equality constraints can be represented in the same syntactic form when they occur in guards or invariants, but the operator itself is not changed.

For a diagnostic trace τ, MPTA-Repair constructs a TARTAR-style trace constraint system [11]. Let $\mathtt{X}\tau$ denote the trace-local variables, including delay variables and clock values at trace positions, and let $\mathtt{V}$ denote the bound variation variables. The bound-varied trace constraint system is written as:

$\mathtt{T}\tau\mathtt{\hat{}bv}(\mathtt{X}\tau, \mathtt{V})$.

It contains the usual constraints for clock initialization, time elapse, clock resets, guard satisfaction, and invariant satisfaction along the trace, except that bounds in guards and invariants are replaced by their varied forms [11]. Bound variation variables affect only the guard and invariant constraints in which the corresponding bound occurs.

For a violated property φ, the local repair obligation for a trace τ is:

$\mathtt{Obl}(\tau, \varphi, \mathtt{V}) = (\exists \mathtt{X}\tau \;.\; \mathtt{T}\tau\mathtt{\hat{}bv}(\mathtt{X}\tau, \mathtt{V})) \wedge (\forall \mathtt{X}\tau \;.\; \mathtt{T}\tau\mathtt{\hat{}bv}(\mathtt{X}\tau, \mathtt{V}) \Rightarrow \Phi\tau,\varphi(\mathtt{X}\tau))$.

The first conjunct requires that the same discrete trace skeleton remains executable after repair. This does not require the same delays or clock valuations as in the original counterexample; it only requires that the same sequence of discrete transitions still has at least one feasible timing realization. The second conjunct requires that no feasible timing realization of this trace skeleton still violates the considered property. Thus, the local repair obligation does not block the functional path itself. It blocks the violating timing realizations of that path.

For a counterexample pool $\mathtt{C}$, where each element is a property-trace pair (τ, φ), MPTA-Repair builds a joint hard constraint:

$\mathtt{FH} = \wedge(\tau,\varphi)\in\mathtt{C}\ \mathtt{Obl}(\tau, \varphi, \mathtt{V}) \wedge \mathtt{WF(V)} \wedge \mathtt{Blocked(V)}$.

Here, $\mathtt{WF(V)}$ contains well-formedness constraints such as non-negative repaired bounds and consistency between lower and upper timing limits when both are present. $\mathtt{Blocked(V)}$ excludes repair assignments that have already failed full-property regression or admissibility. To prefer small repairs, the MaxSMT soft constraints are:

$\mathtt{FS} = \wedge\mathtt{v}\in\mathtt{V}\ (\mathtt{v} = 0)$.

Maximizing the satisfied soft constraints minimizes the number of changed clock bounds [22]. When needed, additional optimization objectives minimize the total magnitude of the nonzero variations. This encoding generalizes single-trace clock-bound repair from one TDT to a pool of currently relevant property-trace obligations.

If a candidate assignment ρ is rejected, MPTA-Repair adds a blocking constraint that prevents the same assignment from being returned again. For the active variation variables $\mathtt{V}\rho$ considered in that candidate, the exact blocking constraint can be written as:

$\mathtt{Block}\rho\mathtt{(V)} = \vee\mathtt{v}\in\mathtt{V}\rho\ \mathtt{v} \neq \rho(\mathtt{v})$.

This does not claim that all similar assignments are invalid. It only prevents the refinement loop from repeating the same failed candidate.

C. Relation-Guided Search Optimization

As refinement proceeds, the number of violated properties, diagnostic traces, and editable clock bounds may grow [23]. Encoding every possible obligation and every editable bound without organization can make the MaxSMT problem larger and may lead to repeated solving over redundant evidence. MPTA-Repair therefore uses relation analysis as a search optimization. These relations guide pruning, grouping, and ranking, but every accepted repair is still validated by full-property regression and admissibility.

At the property layer, MPTA-Repair handles safety properties in a practical normalized fragment. A property is nor-

malized into an A[]-style condition over location predicates and clock or integer comparisons [9]. The relation analysis currently uses conservative syntactic and formula-level rules. Two properties are treated as equivalent when their normalized forms are identical. A dominance relation is used only when it can be established safely. For example, for two upper-bound properties that are identical except for the numerical constant, A[] (Target $\Rightarrow$ x $\leq$ b1) and A[] (Target $\Rightarrow$ x $\leq$ b2),

the first property dominates the second if b1 $\leq$ b2. For lower-bound properties, the direction is reversed. Similar rules apply only when the target predicate, clock, and operator pattern match after normalization. Shared clocks, shared target locations, and shared templates are recorded as dependency information, but they are not by themselves used to remove obligations.

At the counterexample layer, MPTA-Repair maintains a pool of TDTs associated with violated properties. Exact duplicates are removed, and traces are annotated with their involved transitions, locations, clocks, guard bounds, invariant bounds, and source properties [11]. Structural similarity, shared prefixes, and shared repair variables are used to organize the pool and explain why certain traces should be considered together. However, a trace is removed from the active obligation set only when it is an exact duplicate or when a sound subsumption check shows that its local obligation is covered by another encoded obligation. Otherwise, the trace remains available for joint encoding.

At the repair-bound layer, MPTA-Repair maps each property-trace obligation to the clock-bound values that can affect it. This mapping helps restrict the active repair variables to those appearing in or influencing the current counterexample pool. It also helps order candidate generation when several bounds are available. However, relation scores do not prove correctness, and they do not force the solver to modify a particular bound. They only reduce the search effort by avoiding unrelated repair variables when the current counterexample evidence gives no reason to open them.

D. Optimization Objective

The MaxSMT objective is intentionally simple [25]. The primary objective is to minimize the number of changed clock bounds by maximizing soft constraints of the form v = 0. Among candidates with the same number of changed bounds, the implementation may minimize the total magnitude of the changes. This follows the engineering goal of producing small and inspectable repairs.

MPTA-Repair does not optimize directly for speculative notions such as semantic risk or causal relevance. Relation analysis is used before and around solving to select or rank repair variables and obligations. The solver objective itself remains centered on minimality of the edit set and, when enabled, magnitude of the edit values.

Static consistency constraints are enforced during candidate generation [26]. Repaired bounds must remain in the supported numerical domain. When a lower and an upper bound jointly constrain the same clock in a local region, the repaired values must not make the timing interval inconsistent. Since operators and clock references are fixed in the current repair space, generated assignments change only numerical bound values [14].

E. Validation, Admissibility, and Refinement

Candidate validation is the main correctness boundary of MPTA-Repair [27]. After applying a candidate edit set, the repaired model is verified against every property in Φ. A candidate that repairs the triggering property but violates another property is rejected. Such a failure means that the current encoded trace pool was insufficient: the candidate satisfied all encoded local obligations, but the repaired model still has an execution that violates the complete specification. MPTA-Repair therefore collects the newly observed diagnostic trace and adds it to the pool for the next refinement round.

After full-property regression succeeds, MPTA-Repair applies the TARTAR-style admissibility test based on functional equivalence [11]. This check is needed because a clock-bound edit may satisfy safety properties by preventing intended behavior from occurring. TARTAR observes that timed-language equivalence is not a suitable admissibility criterion, because a successful timing repair is expected to remove at least some violating timed realizations [11]. Instead, functional equivalence compares the untimed languages of the original and repaired models [28]. Intuitively, the repaired model should not add or remove functional action traces, even though the timing of those traces may change.

In implementation, this criterion is checked by constructing untimed automata from the original and repaired timed models and comparing their accepted untimed languages [29]. If the languages differ, the repair is rejected and a separating behavior can be reported when available. Only candidates that pass both full-property regression and functional-equivalence admissibility are accepted.

The refinement loop therefore combines local repair generation with global validation. Local TDT encodings generate candidate bound assignments; full-property regression rejects candidates that do not satisfy the complete specification; admissibility rejects candidates that satisfy the properties by changing functional behavior; and blocking constraints prevent repeated failed assignments. The final result is a repair that satisfies the currently stated multi-property specification while preserving the intended untimed behavior of the model.

IV. Implementation and Experimental Evaluation

This section details the implementation of the MPTA-Repair framework and systematically evaluates its practical effectiveness across both standardized benchmark models and complex industrial case studies. To clearly distinguish our targeted contribution while maintaining practical applicability, our methodology strictly focuses on the automated repair of numerical clock bounds within transition guards and location invariants, deliberately avoiding arbitrary alterations to the underlying discrete transition logic. This focused parametric

repair space is pragmatic for industrial contexts, because modifying clock bounds can correct clerical modeling mistakes and provide useful insights for redimensioning system time resources without disrupting the intended functional design. Furthermore, we compare MPTA-Repair with a specialized iterative baseline technique constructed by extending the foundational TARTAR repair idea [12] with a sequential repair loop. This comparison demonstrates the necessity of incorporating multi-property regression and multi-counterexample analysis to prevent secondary errors during repair.

A. Prototype Implementation

We implemented MPTA-Repair as a highly integrated, incremental repair pipeline built around a specialized timed-automata model parser, a verification interface using UPPAAL [19], a MaxSMT-based repair backend powered by Z3 [26], a dedicated admissibility checker, and an overarching iterative refinement control loop. Initially, the repair frontend parses the faulty model to extract editable clock-bound variables embedded within transition guards and location invariants. The trace analyzer then maps observed diagnostic traces to these extracted variables. Concurrently, the relation analyzer establishes dependencies among the specified properties, generated traces, and localized repair sites in order to prioritize candidate generation and reduce the computational search space.

Once the MaxSMT backend synthesizes a viable candidate assignment, the model writer instantiates the corresponding repaired model. Final candidate acceptance relies on two consecutive validation gates designed to guarantee both timing correctness and behavioral fidelity. The first gate performs full-property regression verification through UPPAAL to ensure that no collateral property violations occur. The second gate is an admissibility check for functional equivalence following the TARTAR-style criterion [11]. This module constructs untimed automata from both the original and repaired models and performs a comparative language-equivalence test. The system rejects any candidate that inadvertently adds or removes functional untimed behavior. Whenever either validation gate rejects a candidate, the control loop captures the newly exposed diagnostic traces and feeds them back into the relation analyzer. This initiates an incremental, counterexample-guided refinement iteration that further constrains the MaxSMT solver with updated evidence until a globally correct and admissible repair is found or the search budget is exhausted.

B. Evaluation Strategy and Datasets

We evaluate the proposed framework on two dataset groups. The benchmark dataset consists of timed-automata models collected from UPPAAL and XTA benchmark sources. These models provide structural diversity and allow comparison with existing clock-bound repair settings. The industry dataset contains practical case studies involving railway alarm and communication scenarios [3], [17], train-gate behavior [18], automotive gear control [4], cardiac pacing [6], autonomous road-junction rules [5], mechanical ventilation [7], and aerospace scheduling [8]. Together, the two dataset groups test both algorithmic behavior on established examples and practical behavior on industrial models with synchronization, multiple components, and domain-level timing requirements.

Because naturally occurring faulty timed-automata models are scarce, we use a mutation-based evaluation strategy [24]. Faults are seeded into verified reference models by modifying clock-bound constants in guards and invariants. For simple mutants, one bound is changed and the resulting model is retained if it violates at least one selected timing safety property. For complex mutants, multiple clock-bound edits are injected to simulate interacting timing faults. Generated mutants are filtered to remove invalid or trivial cases: each retained mutant must be syntactically valid and must violate at least one selected property.

We compare MPTA-Repair with an adapted iterative TARTAR-style baseline [12]. For fairness, the baseline receives the same full-property regression and admissibility validation gates used by MPTA-Repair. However, the baseline generates repairs from one violated property or diagnostic trace at a time. It does not jointly analyze relations among multiple properties, counterexamples, and repair variables before candidate generation. This difference allows us to measure the benefit of the multi-property and multi-counterexample repair strategy.

TABLE I
OVERALL REPAIR OUTCOMES REPORTED IN THE SOURCE DRAFT.

Dataset group	Method	Cases	Repaired	Success rate
Benchmark	Iterative TARTAR-style baseline	N1	R1	72%
Benchmark	MPTA-Repair	N1	R2	93%
Industry	Iterative TARTAR-style baseline	N2	R3	20%
Industry	MPTA-Repair	N2	R4	76%

C. Experimental Results

The experimental results show that MPTA-Repair outperforms the iterative baseline, particularly on complex industrial models. On the benchmark dataset, the baseline repairs 72% of the cases, while MPTA-Repair repairs 93%. On the industry dataset, the gap is larger: the baseline repairs 20% of the cases, while MPTA-Repair repairs 76%. These results suggest that the proposed workflow remains effective on established timed-automata benchmarks and is more robust when properties and traces interact in industrial models.

The baseline remains competitive on simpler benchmark models where faults are highly localized. In such cases, a single diagnostic trace often provides enough evidence to infer a useful clock-bound edit. However, in industrial models with interacting concurrent components, synchronization mechanisms, and observer templates, local baseline repairs frequently fail later global validation checks.

Our analysis identifies two main reasons for the stronger performance of MPTA-Repair. First, multi-counterexample accumulation reduces overfitting to one timed diagnostic trace.

A single trace usually exposes only one timing realization of a deeper fault, so a candidate that eliminates that trace may still leave related violations feasible. By accumulating diagnostic traces from multiple operational modes, MPTA-Repair can synthesize small joint edit sets that address the shared timing fault without requiring excessive refinement rounds.

Second, relation-guided optimization improves search efficiency and candidate quality. Property relations help avoid repeatedly solving equivalent obligations. Counterexample relations identify traces that share path fragments or repair variables. Repair-variable relations focus the solver on clock bounds close to the observed violations rather than opening all syntactically editable bounds in the model. In the ablation analysis, removing relation guidance increased the number of generated candidates and the frequency of regression failures. Successful repairs typically modified only one or two clock bounds, and the average edit magnitude remained close to the injected fault size, which makes the suggested repairs easier for engineers to inspect.

D. Discussion and Practical Lessons

The evaluation highlights several practical lessons for applying automated repair to industrial timed automata. First, full-property regression testing is necessary. Industrial properties are often interdependent: a local fix that satisfies a response monitor can violate a separate safety monitor that constrains maximum waiting time. MPTA-Repair avoids accepting such regressions by treating restoration of the full property set as a global objective rather than as a sequence of isolated local patches.

Second, admissibility changes the meaning of repair success. Without checking functional equivalence, a solver could satisfy a timed property by disabling intended behavior, for example by blocking a necessary communication response or preventing a critical action. Such a candidate may eliminate the observed timed diagnostic trace, but it is not a valid repair for an industrial model. The admissibility check therefore prevents property satisfaction from being achieved through unacceptable loss of untimed behavior.

Finally, the unrepaired cases show the limits of the current repair space. Some attempts fail because of solver, verification, or admissibility timeouts, and some models use syntax outside the supported fragment. Other cases appear to lack a feasible repair within numerical clock-bound edits alone. These failures suggest that broader repair actions, such as changing synchronization labels, clock resets, or discrete data updates, are an important direction for future work.

V. Threats to Validity

Internal validity. The prototype includes parsers for timed-automata models, property files, diagnostic traces, repair sites, and admissibility artifacts. Bugs in parsing, trace reconstruction, SMT encoding, candidate application, or admissibility checking may affect the results. We mitigate this threat by validating every accepted candidate using the external model checker and by recording repair reports for each run.

Construct validity. The evaluation uses injected clock-bound faults. Such faults are appropriate for evaluating clock-bound repair, but they do not cover all real modeling errors. Real faults may involve wrong resets, missing transitions, incorrect synchronizations, data updates, or erroneous properties. The results should therefore be interpreted as evidence for clock-bound repair, not for arbitrary timed-automata repair.

External validity. Although the industry dataset covers several practical domains, it is still a finite set of open or reconstructed models. It may not represent all industrial timed-automata models. Larger proprietary models, richer data variables, and more complex property specifications may introduce additional challenges.

Baseline fairness. The baseline is an iterative TARTAR-style repair workflow adapted to the multi-property setting by adding full-property regression and admissibility checking. This makes it stronger than a one-shot baseline, but it is still not identical to the original setting for which TARTAR was designed. We therefore interpret the comparison as a comparison against a strong single-property repair strategy, not as a claim that TARTAR itself is unsuitable in general.

Property selection. Industry-case properties were selected from model descriptions, papers, embedded queries, and intended timing requirements. Some properties may be incomplete or may not capture all domain-level requirements. We mitigate this by requiring all reference models to satisfy the selected property set and by filtering out trivial guard-mirror properties when constructing the benchmark.

Repair-space limitation. The current method focuses on numerical clock-bound edits in guards and invariants [11]. Some failed cases likely require repair actions outside this space. Extending the method to reset repair, synchronization repair, clock-reference repair, and structural repair is left for future work.

VI. Related Work

The closest related work is clock-bound repair for timed systems and TarTar [12]. Clock-bound repair encodes timed diagnostic traces into linear real arithmetic and uses MaxSMT solving to compute small modifications to guard and invariant bounds [11]. TarTar extends this line into a repair analysis tool and introduces an admissibility check based on functional equivalence. MPTA-Repair builds on this foundation, but targets a different setting: multiple properties, multiple diagnostic traces, and full-property validation in timed-automata models from practical industrial cases.

A second related direction is parameter synthesis for parametric timed automata [13]. Tools such as IMITATOR synthesize parameter valuations that satisfy timing constraints or preserve behavior around a reference valuation [30]. These techniques are powerful for design-space exploration and robustness analysis [15]. MPTA-Repair differs in its use of diagnostic traces and its focus on producing small syntactic repairs for faulty models rather than synthesizing a full parameter region.

A third related direction is model repair and automated program repair using SAT, MaxSAT, SMT, and counterexample-guided refinement [31]. These works provide search and optimization ideas such as fault localization [16], candidate blocking [32], repair minimization [33], and abstraction refinement [23]. MPTA-Repair adapts this style of reasoning to the semantics of timed automata, where candidate validity must consider both timing properties and untimed behavior preservation.

Finally, this work is related to industrial verification and dependability engineering [34]. Industry-oriented verification often emphasizes full regression, explainability, traceability, and lessons learned rather than only theoretical completeness [35]. MPTA-Repair follows this perspective by treating repair as an engineering workflow: diagnostic traces generate candidates, but acceptance depends on full regression evidence and admissibility.

VII. Conclusion

This paper presented **MPTA-Repair**, an automatic repair workflow for multi-property, multi-counterexample clock-bound repair of industrial timed-automata models. The method addresses a practical limitation of local TDT-based repair: industrial models are commonly validated against multiple timing properties, and one timing defect may produce several related counterexamples. Repairing one trace in isolation may therefore be insufficient.

MPTA-Repair collects diagnostic traces from violated properties, uses lightweight relations among properties, traces, and repairable bounds to guide repair search, and synthesizes small clock-bound edits using a MaxSMT-based backend. Every candidate is validated by full-property regression and a TARTAR-style admissibility check based on functional equivalence [11]. This design ensures that accepted repairs are not merely local counterexample fixes, but system-level timing corrections that preserve intended behavior.

Our evaluation on benchmark and industry timed-automata datasets shows that MPTA-Repair improves repair success over an iterative TARTAR-style baseline, especially on industry cases where multi-property interaction and admissibility constraints are prominent. The results also show that failures remain meaningful: unrepaired cases often require changes outside the clock-bound repair space or exceed the configured solving and validation budgets.

Future work will extend the repair space beyond numerical clock bounds to include resets, clock references, synchronization labels, and structural edits. We also plan to strengthen counterexample generation, improve relation-guided decomposition, and evaluate the method on larger industrial timed-automata models.

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